

CS5275 The Algorithm Designer's Toolkit
S2 AY2025/26
Problem set 3

Submission is due at 23:59, March 19 (Thu).

Problem 1 (Staying inside a low-conductance set, 4 marks). Let $G = (V, E)$ be a graph and let $(S, V \setminus S)$ be a cut with conductance $\Phi(S) = \phi$. Show that there exists a subset $X \subseteq S$ such that

$$\text{vol}(X) \geq 0.8 \cdot \text{vol}(S),$$

and for every $v \in X$,

$$\Pr[\text{a } t\text{-step standard (non-lazy) random walk started at } v \text{ never leaves } S] \geq \frac{1}{2},$$

where

$$t = \left\lfloor \frac{0.1}{\phi} \right\rfloor.$$

Hint: The intended solution builds on the proof idea used in Tutorial Problem Set 2. You may draw on ideas discussed during the tutorial, but you should still present a complete, formal proof. Specifically, you cannot simply use the results in that problem set, you need to prove any result you use from there.

Problem 2 (Maximum cut of Ramanujan graphs, 4 marks). Let $G = (V, E)$ be any n -vertex d -regular Ramanujan graph. The goal of the problem is to show that the size of a maximum cut is

$$\frac{nd}{4} \pm O(n\sqrt{d}).$$

2.a (2 marks) Prove that for any cut $(S, V \setminus S)$,

$$|E(S, V \setminus S)| \leq \frac{nd}{4} + O(n\sqrt{d}).$$

2.b (2 marks) Prove that there exists a cut $(S, V \setminus S)$ such that

$$|E(S, V \setminus S)| \geq \frac{nd}{4} - O(n\sqrt{d}).$$

Problem 3 (Vertex connectivity of regular expanders, 4 marks). Let $G = (V, E)$ be a connected d -regular graph with conductance $\Phi(G) \geq \phi$.

Prove that the vertex connectivity $\kappa(G)$ satisfies

$$\kappa(G) \in \Omega(\phi d).$$

Problem 4 (Finding a minimum vertex cut with a promise, 4 marks). The goal of this problem is to show that, given the promise that the minimum vertex cut is small, it can be found using fewer than $O(n^2)$ s - t cut queries.

Suppose we have an oracle that computes an s - t minimum vertex cut.

Let $G = (V, E)$ be a graph, and let $k = \lfloor \sqrt{n} \rfloor$. Let S be an arbitrary set of $k + 1$ vertices, and construct a new graph G' by adding vertex r adjacent to every vertex in S .

Design an algorithm that, using at most $O(n)$ oracle queries on G and/or G' , and taking $O(n^2)$ time aside from the oracle calls, either

1. finds a minimum vertex cut of G , or
2. correctly reports that $\kappa(G) > k$.

Prove that your algorithm is correct.